

bio-signal acquisition system detects subtle physiological signals, while a programmable mixed-signal device performs analogue conditioning, reducing the computational load on the microcontroller. The microcontroller processes sensor data, runs health-monitoring algorithms, and efficiently manages system functions.



The design also focuses on power optimisation through careful component selection, low-power firmware, and NFC-based wireless charging that avoids exposed connectors. This approach improves durability, convenience, and device longevity.

By handling sensing, signal processing, wireless communication, and power management within the reference design, development time, firmware effort, and integration risk are reduced for engineers building wearable health-monitoring products.

Applications. The reference design is suitable for a range of applications, including fitness trackers, remote patient monitoring systems, and industrial wellness devices.

Link. <https://www.electronicsforu.com/electronics-projects/low-power-health-monitor-band-reference-design-for-wearables>

OEM Brand. Renesas Electronics Corporation

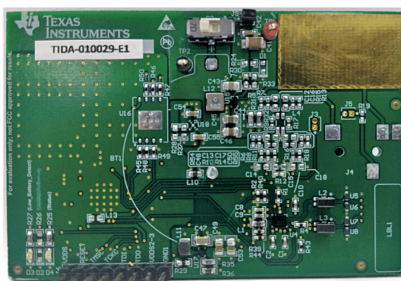
Wearable oxygen and heart rate monitor

This reference design supports the development of wearable optical heart-rate and SpO₂ monitors with support for multiple LEDs and multiple photodiodes, without

requiring additional sensors or complex circuitry.

It acquires phase-shifted PPG signals to improve measurement accuracy, enabling calculation of heart rate, SpO₂, and related physiological parameters. Red and infrared LEDs measure blood oxygen levels, while green LEDs improve heart-rate detection.

The device includes wireless transmission for continuous monitoring, along with basic battery and connectivity indicators. It can operate for up to 30 days on a single coin-cell battery.



By managing LED control, multi-wavelength measurement, signal acquisition, and wireless communication within the reference design, the system reduces design complexity, integration effort, and development risk for engineers building wearable health devices.

Applications. The reference design is suitable for wearable applications.

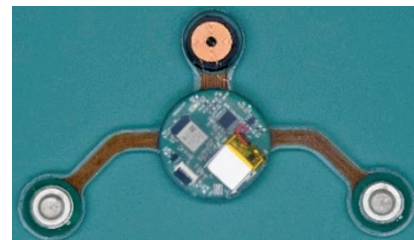
Link. <https://www.electronicsforu.com/electronics-projects/wearable-oxygen-and-heart-rate-monitor-reference-design>

OEM Brand. Texas Instruments (TI)

Wearable ECG systems

This reference design helps you build wearable ECG systems that capture clean heart signals, even during motion, without adding complex correction circuits. It combines ECG, SCG, GCG, and motion sensing in a single electronic skin patch, allowing engineers to correct ECG data using real-time movement information. The system captures

small ECG signals while filtering out high- and low-frequency noise, including motion artifacts, breathing, EMG, and power-line interference. Motion data from an accelerometer and inertial measurement unit can be used with filtering and adaptive algorithms to reduce baseline drift and spikes caused by user activity. The board supports heart rate, heart



rate variability, anomaly detection, and context-aware signal capture, adapting measurements when the user is resting, walking, or exercising. The system is powered by a 35mAh lithium-ion polymer battery, uses silicone encapsulation, provides Bluetooth connectivity for wireless monitoring, and relies on a single-lead ECG setup with two electrodes to reduce size and complexity while enabling practical wearable deployment.

Applications. This reference design is suited for home health monitoring, fitness tracking, telehealth, wearable prototyping, context-aware heart monitoring, and cardiac anomaly detection.

Link. <https://www.electronicsforu.com/electronics-projects/reference-design-for-wearable-ecg-systems>

OEM Brand. STMicroelectronics

These reference designs provide engineers and researchers with ready-to-use platforms for developing wearable health and fitness devices. They integrate sensing, signal processing, wireless communication, and power management within compact, low-power architectures. **EFY**

Nidhi Agarwal is Senior Technology Journalist at EFY with a deep interest in embedded systems, development boards, and IoT cloud solutions.